Chapter 17: Subspaces Worksheet

1. Are each of the following statements true or false?

(a) If S and T are subsets of $\mathbb{R}^2$ such that $\text{span}(S) = \text{span}(T)$, then $S = T$.

(b) If $\mathbf{u} + \mathbf{v}$ is in a subspace U, then $\mathbf{u}$ and $\mathbf{v}$ must also be in U.

(c) If A is a 4×3 matrix, then the solution space of $A\mathbf{x} = \mathbf{0}$ is a subspace of $\mathbb{R}^4$.

(d) The subset of 2×2 matrices $U = \{A | A \text{ in } M_{22}, A^2 = O\}$ is a subspace of M_{22} (note that O denotes the zero matrix).

(e) If U is a subspace of V, then if a, b are any real numbers and $\mathbf{u}, \mathbf{v}$ are in U, $a\mathbf{u} + b\mathbf{v}$ is also in U.

(f) $x^2 + 9x - 11$ lies in $\text{span}\{x^2 - 1, x^2 - 2x + 3, 2x^2 + 3x\}$ (Hint: look at your previous worksheet).

(g) $P_1 = \text{span}\{x + 1, x - 1\}$

2. Consider $\mathbb{R}^3$, and consider the subset W consisting of vectors of the form $\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$, where $3v_1 + v_2 - 2v_3 = 0$, i.e.

$$W = \left\{ \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \middle| 3v_1 + v_2 - 2v_3 = 0 \right\}$$

Follow the steps below to show that W is a subspace of $\mathbb{R}^3$:

(a) Show that the zero vector in $\mathbb{R}^3$ is in the set W. (This shows it's nonempty)

(b) Show that for any two vectors $\mathbf{u}$ and $\mathbf{v}$ in W, $\mathbf{u} + \mathbf{v}$ is in W.

(c) Show that for any real number c and any $\mathbf{u}$ in W, $c\mathbf{u}$ is in W.

3. Each of the following are subsets of $\mathbb{R}^3$, but they are NOT subspaces of $\mathbb{R}^3$. Find explicitthat means examples with numbers, not variables counterexamples to show that each W is not a subspace of $\mathbb{R}^3$. Your work should include a demonstration that your counterexamples violate either condition (2) **[x]**

$$\boxed{\textit{If } \mathbf{u} \textit{ and } \mathbf{v} \textit{ are any vectors in } W,\ \mathbf{u}+\mathbf{v} \textit{ is in } W}$$

or condition (3) **[x]**

$$\boxed{\textit{If } c \textit{ is a real number and } \mathbf{u} \textit{ is any vector in } W, \textit{ then } c\mathbf{u} \textit{ is in } W}$$

from the ***subspace*** definition.

(a) W is the set of vectors of the form $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, where x_1, x_2, x_3 are all even numbers.

(b) W is the set of vectors of the form $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, where $x_1 x_3 \geq 0$.

(c) W is the set of vectors of the form $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, where $x_2 - x_1 = 2$.

4. In each case determine if $\mathbf{x}$ lies in $U = \operatorname{span}(\mathbf{y}, \mathbf{z})$. If it is, write it as a linear combination of $\mathbf{y}$ and $\mathbf{z}$. If it is not, show why not.

(a) $\mathbf{x} = \begin{bmatrix} -1 \\ 4 \\ 2 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} -2 \\ 2 \\ 2 \end{bmatrix} \quad \mathbf{z} = \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}.$

(b) $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 15 \\ 11 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} 2 \\ -1 \\ 0 \\ 2 \end{bmatrix} \quad \mathbf{z} = \begin{bmatrix} 1 \\ -1 \\ -3 \\ 1 \end{bmatrix}.$

5. Bob, a linear algebra student, is trying to answer the question: Is the set of vectors $\begin{bmatrix} x \\ y \end{bmatrix}$ where $x + y \geq 0$ a subspace of $\mathbb{R}^2$? Bob checks that $\begin{bmatrix} -1 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$ are in the set, and $\begin{bmatrix} -1 \\ 3 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$ is in the set. He checks that $2\begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} -2 \\ 6 \end{bmatrix}$ is in the set. Bob then concludes that since the sum of the vectors is in the set, and the scalar multiple of $\begin{bmatrix} -1 \\ 3 \end{bmatrix}$ is in the set, it must be a subspace.

Is Bob's argument correct? If not, how can you fix it?

6. Consider P_3, the vector space of polynomials of degree 3 or less (with standard polynomial addition and scalar multiplication), and consider the following set:

$$W = \left\{ p(x) \text{ in } P_3 \middle| \int_0^1 p(x)dx = 0 \right\}$$

Does W form a subspace of P_3? Completely justify your answer.

7. Consider the set $W = \left\{ \begin{bmatrix} r \\ s \\ 2r+4s \end{bmatrix} \middle| r, s \text{ in } \mathbb{R} \right\}$. Show (using the subspace definition) that W is a proper subspace of $\mathbb{R}^3$. Recall: ***proper*** means that W is a subspace other than $\{\mathbf{0}\}$ or $\mathbb{R}^3$.

8.

(a) If A and B are $m \times n$ matrices, show (using the subspace definition) that $U = \{\mathbf{x} \text{ in } \mathbb{R}^n | A\mathbf{x} = B\mathbf{x}\}$ is a subspace of $\mathbb{R}^n$.

(b) Fill in the blank: If A and B are $m \times n$ matrices, $U = \{\mathbf{x} \text{ in } \mathbb{R}^n | A\mathbf{x} = B\mathbf{x}\}$ is equal to the solution space of __________.

(c) What if A is $m \times n$, B is $k \times n$, and $m \neq k$?

9. Consider U, the set of 2×2 symmetric matrices.

(a) Prove that U is a subspace of $M_{2\times 2}$ using the subspace test.

(b) Show that $S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\}$ is a spanning set for U.

(c) Show that each vector in S is a linear combination of the vectors in $T = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\}$ and show that each vector in T is a linear combination of the vectors in S. This shows that T is also a spanning set for U.

10. Consider $M_{3\times 3}$, the vector space of 3×3 matrices (with standard matrix addition and scalar multiplication), and let $\mathbf{x}$ be a vector in $\mathbb{R}^3$. Let U be the set of matrices A such that $A\mathbf{x} = \mathbf{0}$. Prove that U is a subspace of $M_{3\times 3}$.